Linear Algebra — Lecture 7

Determinants of Order 2, Determinants of Order n, Cofactor Expansion

Friedberg–Insel–Spence, *Linear Algebra* 4e, §4.1–§4.2
D. C. Lay, *Linear Algebra and Its Applications* 4e, §3.1

Topics. Determinants of order 2 and their linearity in each row, the recursive definition of the determinant of order n, minors and cofactors, and cofactor expansion along any row.

How the two texts divide the work. FIS (the main text) develops the theory of an **abstract vector space over an arbitrary field** $\mathbb{F}$; while Lay supplies concrete examples in $\mathbb{R}^n$ and the computational intuition. Mapping:

Topic	Lay 4e (printed)	Friedberg 4e (printed)
Determinants of order 2	§3.1 (p. 164–165)	§4.1 (p. 199–201)
Order 3 and expansion	§3.1 (p. 165–166)	§4.2 (p. 209–211)
Linearity in each row	—	§4.2 (p. 212)
Expansion along any row	Theorem 1 (p. 166)	§4.2 (p. 215)

Two questions that drive the whole lecture: (1) *A square matrix is assigned a single number: by which rules is that number pinned down?* (2) *Cofactor expansion offers n rows to choose from; which choice is cheapest, and how are the signs kept straight?*

How to read: use Lay first to get the expansion of a 3×3 under your fingers, then Friedberg to see why the recursive definition and "expansion along any row" are correct.

All page numbers here are printed pages. Lay: PDF page = printed page +17. Friedberg: PDF page = printed page +15.

1 Determinants of Order 2

The first lectures treated a matrix as a linear transformation and as a system of equations; the matrix itself stayed a table of numbers. From this lecture on, each square matrix receives a single number. Two rules characterise it: fixing the other rows, it is linear in one row, and it changes sign when two rows are interchanged. In order 2 the formula can be written down at once; in order n the determinant is defined recursively through cofactors.

Definition 1.1 (Determinant of order 2; Friedberg §4.1, p. 200)**.** Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ be a 2×2 matrix with entries from a field $\mathbb{F}$. The **determinant** of A, written $\det A$ or $|A|$, is the scalar

$$ad - bc.$$

Example 1.1 (Friedberg Example 1, §4.1, p. 200)**.** Find the determinants of $A = \begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$ and $B = \begin{bmatrix} 3 & 2 \\ 6 & 4 \end{bmatrix}$.

Solution. Both follow the definition:

$$\det A = 1 \cdot 4 - 2 \cdot 3 = -2, \qquad \det B = 3 \cdot 4 - 2 \cdot 6 = 0.$$

In B the second row is twice the first, so the two products agree and the determinant vanishes. □

Approach. Order 2 has a single formula: subtract the product of one diagonal from the product of the other. Proportional rows force the value 0, which is the rank criterion of Lecture 4 seen from the determinant side.

Theorem 1.1 (Linearity in each row; Friedberg Theorem 4.1, §4.1, p. 200)**.** *Let $\boldsymbol{u}, \boldsymbol{v}, \boldsymbol{w} \in \mathbb{F}^2$ and $k \in \mathbb{F}$. With the other row held fixed, the determinant is a linear function of this row:*

$$\det \begin{bmatrix} \boldsymbol{u} + k\boldsymbol{v} \\ \boldsymbol{w} \end{bmatrix} = \det \begin{bmatrix} \boldsymbol{u} \\ \boldsymbol{w} \end{bmatrix} + k \det \begin{bmatrix} \boldsymbol{v} \\ \boldsymbol{w} \end{bmatrix}, \qquad \det \begin{bmatrix} \boldsymbol{w} \\ \boldsymbol{u} + k\boldsymbol{v} \end{bmatrix} = \det \begin{bmatrix} \boldsymbol{w} \\ \boldsymbol{u} \end{bmatrix} + k \det \begin{bmatrix} \boldsymbol{w} \\ \boldsymbol{v} \end{bmatrix}.$$

Proof. Write $\boldsymbol{u} = (a_1, a_2)$, $\boldsymbol{v} = (b_1, b_2)$, $\boldsymbol{w} = (c_1, c_2)$. By definition the left side is

$$(a_1 + kb_1)c_2 - (a_2 + kb_2)c_1 = (a_1c_2 - a_2c_1) + k(b_1c_2 - b_2c_1),$$

which is the sum of the two determinants on the right. The second identity follows the same way with c moved to the top row. □

Remark 1.1 (det is not a linear transformation; Friedberg §4.1, p. 200)**.** The two matrices of Example 1.1 add up to $A + B = \begin{bmatrix} 4 & 4 \\ 9 & 8 \end{bmatrix}$, so $\det(A + B) = 4 \cdot 8 - 4 \cdot 9 = -4$ while $\det A + \det B = -2 + 0 = -2$. The two differ, so $\det : M_{2\times 2}(\mathbb{F}) \to \mathbb{F}$ is not a linear map: it is linear along each row separately, not on matrices as a whole.

Remark 1.2 (The invertibility criterion of order 2)**.** Lecture 4 notes, Theorem 3.1 already states that $\det A \neq 0$ exactly when A is invertible, and gives $A^{-1} = \frac{1}{\det A} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$. Friedberg Theorem 4.2 (§4.1, p. 201) says the same thing with a rank argument: when $\det A \neq 0$ one verifies directly that the displayed matrix is the inverse; when $\det A = 0$ the rank of A is at most 1.

2 Determinants of Order n and Cofactor Expansion

The formula for order 2 has no ready-made counterpart in order n, but deleting one row and one column leaves a smaller square matrix, and its determinant can be taken in turn. The determinant is therefore defined by recursion.

Definition 2.1 (Minor, cofactor, determinant of order n; Friedberg §4.2, p. 209–210)**.** Let $A \in M_{n\times n}(\mathbb{F})$ with $n \geq 2$. Deleting row i and column j of A leaves an $(n-1) \times (n-1)$ matrix, written $\tilde{A}_{ij}$ and called the **minor** of A_{ij}. The scalar

$$c_{ij} = (-1)^{i+j} \det(\tilde{A}_{ij})$$

is the **cofactor** of A_{ij}. The determinant of A is defined recursively by **cofactor expansion along the first row**:

$$\det A = \sum_{j=1}^{n} (-1)^{1+j} A_{1j} \det(\tilde{A}_{1j}) = A_{11}c_{11} + A_{12}c_{12} + \cdots + A_{1n}c_{1n}.$$

Example 2.1 (Friedberg §4.2, Example 1, p. 210)**.** Find the determinant of $A = \begin{bmatrix} 1 & 3 & -3 \\ -3 & -5 & 2 \\ -4 & 4 & -6 \end{bmatrix}$.

Solution. Expand along the first row; the three minors are

$$\det \begin{bmatrix} -5 & 2 \\ 4 & -6 \end{bmatrix} = 30 - 8 = 22, \quad \det \begin{bmatrix} -3 & 2 \\ -4 & -6 \end{bmatrix} = 18 + 8 = 26, \quad \det \begin{bmatrix} -3 & -5 \\ -4 & 4 \end{bmatrix} = -12 - 20 = -32.$$

The signs alternate as $(-1)^{1+j}$, hence

$$\det A = 1 \cdot 22 - 3 \cdot 26 + (-3) \cdot (-32) = 22 - 78 + 96 = 40. \qquad \square$$

Approach. First turn each minor into a single number, then combine them with alternating signs. A sign depends only on the position (i, j), never on the sign of the entry itself; here $(-1)^{1+j}$ gives the checkerboard pattern plus, minus, plus.

Theorem 2.1 (Linearity in each row; Friedberg Theorem 4.3, §4.2, p. 212)**.** *Let $A \in M_{n\times n}(\mathbb{F})$. If row r is replaced by $\boldsymbol{u} + k\boldsymbol{v}$ while the other rows stay fixed, the determinant equals the sum of the determinants obtained by putting $\boldsymbol{u}$ and $\boldsymbol{v}$ in row r, the second weighted by k.*

Proof. Induction on n. For $n = 1$ the determinant is the entry itself, so the claim is clear. Let $n \geq 2$ and assume the result for order $n - 1$.

If $r = 1$: by definition $\det A = \sum_j A_{1j} c_{1j}$, and the cofactor c_{1j} involves only rows 2 to n, so the sum is linear in row 1.

If $r > 1$: expand along the first row, $\det A = \sum_j A_{1j} c_{1j}$ with $c_{1j} = \pm \det(\tilde{A}_{1j})$. Row $r - 1$ of $\tilde{A}_{1j}$ is row r of A and its other rows agree with those of A; by the induction hypothesis each $\det(\tilde{A}_{1j})$ is linear in that row, so the weighted sum of these determinants of order $n - 1$ is linear in row r of A. $\square$

Lemma 2.2 (Friedberg §4.2, p. 213)**.** *Let $B \in M_{n\times n}(\mathbb{F})$ with $n \geq 2$. If row i of B is $\boldsymbol{e}_k$, then*

$$\det B = (-1)^{i+k} \det(\tilde{B}_{ik}).$$

Proof. For $i = 1$ the definition gives the claim at once: the first row has 1 only in position k, so a single term survives, $(-1)^{1+k} \cdot 1 \cdot \det(\tilde{B}_{1k})$. The case $i > 1$ is proved by induction on n: expand B along its first row and apply the induction hypothesis to each minor of order $n - 1$; the signs combine to $(-1)^{i+k}$. The step-by-step details are in Friedberg §4.2, p. 213–215. $\square$

Theorem 2.3 (Expansion along any row; Friedberg Theorem 4.4, §4.2, p. 215)**.** *Let $A \in M_{n\times n}(\mathbb{F})$ and $1 \leq i \leq n$. Then*

$$\det A = \sum_{j=1}^{n} (-1)^{i+j} A_{ij} \det(\tilde{A}_{ij}).$$

Proof. Write row i as $A_{i1}\boldsymbol{e}_1 + \cdots + A_{in}\boldsymbol{e}_n$. Applying Theorem 2.1 to row i repeatedly gives

$$\det A = \sum_{j=1}^{n} A_{ij} \det(B_j),$$

where B_j is obtained from A by replacing row i with $\boldsymbol{e}_j$. By Lemma 2.2,

$$\det(B_j) = (-1)^{i+j} \det(\tilde{A}_{ij}),$$

and the claim follows. $\square$

Remark 2.1 (Why only rows for now)**.** Theorem 1 of Lay (§3.1, p. 166) states the stronger fact that expansion may follow any row or any column. The column half rests on $\det(A^\top) = \det A$; that identity and the remaining properties of the determinant are the subject of Lecture 9.

Remark 2.2 (The $n!$-term formula and inversions)**.** Unfolding the cofactor expansion layer by layer writes the determinant as a sum of $n!$ terms. Let σ be a permutation of $\{1, \ldots, n\}$ and let $\tau(\sigma)$ be its **number of inversions**, the count of pairs $i < j$ with $\sigma(i) > \sigma(j)$; put $\operatorname{sgn}(\sigma) = (-1)^{\tau(\sigma)}$. Then

$$\det A = \sum_{\sigma} \operatorname{sgn}(\sigma)\, A_{1\sigma(1)} A_{2\sigma(2)} \cdots A_{n\sigma(n)},$$

the sum running over all $n!$ permutations. For $n = 2$ the two terms are $A_{11}A_{22} - A_{12}A_{21}$, which agrees with Definition 1.1; for $n = 3$ the six terms combine into the expansion one remembers by drawing diagonals.

This formula and cofactor expansion produce the same number: each step of the recursion takes one entry from a row and deletes its column, and after n steps what has been taken is exactly a permutation; the sign is the parity of the number of interchanges needed to sort that permutation back to $1, 2, \ldots, n$, which is $(-1)^{\tau(\sigma)}$.

Neither textbook adopts this definition. Friedberg defines the determinant recursively through cofactors (§4.2, p. 209–210) and then, in §4.5* (from p. 238), characterises it as the unique function that is linear in each row and vanishes on matrices with two equal rows; Lay expands along a row (§3.1, p. 165–166). The inversion formula is equivalent to those characterisations and merely writes the unfolding out as an explicit sum.

3 A Worked Example

Example 3.1 (Expanding along the row with the most zeros; written for these notes; method from Friedberg §4.2, p. 215)**.** Find the determinant of

$$A = \begin{bmatrix} 1 & 2 & 0 & 3 \\ 0 & 0 & 5 & 0 \\ 2 & 1 & 1 & 4 \\ 3 & 0 & 0 & 2 \end{bmatrix}.$$

Solution. Row 2 has a single nonzero entry, so expanding along it is cheapest. With $A_{23} = 5$,

$$\det A = (-1)^{2+3} \cdot 5 \cdot \det(\tilde{A}_{23}) = -5 \det \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 4 \\ 3 & 0 & 2 \end{bmatrix}.$$

The remaining determinant of order 3 is again expanded along its row with the most zeros, row 3:

$$\det \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 4 \\ 3 & 0 & 2 \end{bmatrix} = 3 \cdot (-1)^{3+1} \det \begin{bmatrix} 2 & 3 \\ 1 & 4 \end{bmatrix} + 2 \cdot (-1)^{3+3} \det \begin{bmatrix} 1 & 2 \\ 2 & 1 \end{bmatrix} = 3 \cdot 5 + 2 \cdot (-3) = 9.$$

Hence $\det A = -5 \cdot 9 = -45$.

Check: expansion along the first row of the same matrix gives the minors

$$\det \begin{bmatrix} 0 & 5 & 0 \\ 1 & 1 & 4 \\ 0 & 0 & 2 \end{bmatrix} = -10, \qquad \det \begin{bmatrix} 0 & 5 & 0 \\ 2 & 1 & 4 \\ 3 & 0 & 2 \end{bmatrix} = 40, \qquad \det \begin{bmatrix} 0 & 0 & 5 \\ 2 & 1 & 1 \\ 3 & 0 & 0 \end{bmatrix} = -15,$$

so $1 \cdot (-10) - 2 \cdot 40 + (-1) \cdot 3 \cdot (-15) = -10 - 80 + 45 = -45$, in agreement with the value above. □

Approach. Count the nonzero entries in each row and expand along the sparsest one; one step takes the 4×4 down to order 3, and the same rule is applied again. The sign depends on the position (i, j): here $(-1)^{2+3} = -1$, $(-1)^{3+1} = +1$ and $(-1)^{3+3} = +1$, and each must be read off separately rather than carried over from the previous row.

Example 3.2 (A determinant with a parameter; written for these notes)**.** Let

$$p(x) = \det \begin{bmatrix} x & 1 & 1 \\ 1 & x & 1 \\ 1 & 1 & x \end{bmatrix}.$$

Find $p(x)$ and solve $p(x) = 0$.

Solution. Expanding along the first row,

$$p(x) = x \det \begin{bmatrix} x & 1 \\ 1 & x \end{bmatrix} - \det \begin{bmatrix} 1 & 1 \\ 1 & x \end{bmatrix} + \det \begin{bmatrix} 1 & x \\ 1 & 1 \end{bmatrix} = x(x^2 - 1) - (x - 1) + (1 - x).$$

Collecting terms,

$$p(x) = x^3 - x - x + 1 + 1 - x = x^3 - 3x + 2 = (x - 1)^2(x + 2),$$

so $p(x) = 0$ at $x = 1$ (double) and $x = -2$. □

Approach. At $x = 1$ the three rows coincide, so the determinant must vanish and $x - 1$ is a factor; dividing it out leaves a quadratic, which is cheaper than solving a cubic from scratch.

4 Checklist and Exercises

By the end of this lecture you should be able to:

1. state the definition of a determinant of order 2 and explain why it is linear in each row but not linear on matrices as a whole;
2. define the minor and the cofactor, and write the recursive definition of the determinant of order n;
3. explain why expansion along any row gives the same number as expansion along the first row, naming the theorems used;
4. decide the sign of each term in an expansion, and explain why the checkerboard $(-1)^{i+j}$ does not depend on the sign of the entry;
5. pick the sparsest row of a 4×4 matrix and expand along it to reduce the problem to order 3.

Exercises (printed pages). Lay: §3.1 p. 167 (Exercises 1–8: expansion along the first row; 9–14: expansion along any row or column; 19–24: how elementary row operations affect a determinant); Friedberg: §4.1 p. 207 (Exercises 1–9: determinants of order 2 and their linearity), §4.2 p. 220 (Exercises 1–14: cofactor expansion and determinants of order n). Each section is preceded by Practice Problems; do those first.

How to use the two texts together:

- **Computation and proof divide cleanly.** How a 3×3 is expanded, which row to pick, how the signs are decided: Lay (§3.1). Why the determinant can be defined recursively and why the choice of row does not matter: Friedberg (§4.2).
- The spine of the lecture: *the formula for order 2 → linearity in each row → minors and cofactors → the recursive definition along the first row → expansion along any row.*
- The one result worth keeping is Theorem 2.3: it widens "expand along the first row", which is all the definition offers, into "expand along any row", and that is what makes a 4×4 or larger tractable. Its proof uses only linearity in each row (Theorem 2.1) and Lemma 2.2.